Chit Fund Project - Payment Module Notes

Complete Architecture, Database Schema, FastAPI APIs & React Frontend Guide

1. System Architecture & Workflow

The Payment Module operates on a **2-Step Verification Model** to ensure secure transactions and valid payment receipts:

- **Step 1: Member Views Dues** - Member checks active monthly dues via GET /api/v1/member/payments/dues.
- **Step 2: Emergency Reason & QR Scan** - Member clicks *Pay via Instant UPI / QR*, selects emergency reason (Rank 1-5), and scans GPay QR Code.
- **Step 3: Request Submission** - Calls POST /razorpay/verify. Payment record created with status **pending**, installment status updated to **paid**.
- **Step 4: Admin Verification** - Admin reviews transaction ID and UPI ID in Admin Panel. Clicking Approve sets payment status to **success** and installment to **verified**.
- **Step 5: Verified Receipt** - Formal receipt (REC-2026-XXXX) is issued to member with confetti celebration.

2. Database Schema (SQLAlchemy)

Field Name	Data Type	Description
id	Integer (PK)	Unique payment identifier
installment_id	Integer (FK)	References installments.id
user_id	Integer (FK)	References users.id
amount	Float	Monthly installment payment amount
payment_method	String(50)	e.g. Google Pay UPI (nareshk1213@oksbi)
transaction_id	String(100)	Unique UPI payment / transaction ID
status	Enum	pending | success | failed
notes	Text	Emergency reason & Admin notes

3. FastAPI Backend Endpoints

Member Router (/api/v1/member/payments):

- GET /dues - List all unpaid installments for current member.
- POST /razorpay/create-order - Generate Razorpay Order ID for test mode checkout.
- POST /razorpay/verify - Submit member payment request for Admin verification.
- GET /history - Retrieve member payment receipt history.

Admin Router (/api/v1/admin/payments):

- GET / - List all payments with filter by status (pending/success/failed).

- PUT /{id}/status - Update status (success -> marks installment verified).
- POST /send-due-reminders - Trigger monthly due notification reminders.

4. React Frontend Components

- **Payments.jsx**: Displays active dues & past transaction history with status badges.
- **RazorpayModal.jsx**: Modal featuring GPay QR code, UPI copy, emergency reason dropdown, and polling.
- **PaymentReceiptModal.jsx**: Modal displaying official verified digital receipt with print capability.